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Inventor(s)

Diep; Vinh Quang et al.

SELECTIVE INCREASE AND DECREASE TO PASS VOLTAGES FOR PROGRAMMING A MEMORY DEVICE

Abstract

A memory device includes a memory array with a plurality of wordlines and control logic operatively coupled with the memory array. The control logic determines, during an initial phase of a program operation, whether a program voltage being applied to a selected wordline of the memory array satisfies a threshold program voltage. The control logic increases, in response to the program voltage not satisfying the threshold program voltage, an initial pass voltage to generate a higher pass voltage. The control logic causes the higher pass voltage to be applied to unselected wordlines of the plurality of wordlines before programming one or more memory cells associated with the selected wordline.

Inventors: Diep; Vinh Quang (Hayward, CA), Tsai; Jeffrey Ming-Hung (San Jose, CA), Lu; Ching-Huang (San Jose, CA), Dong; Yingda (Los Altos, CA)

Applicant: Micron Technology, Inc. (Boise, ID)

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Background/Summary

REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/103,978, filed Jan. 31, 2023, which claims the benefit of U.S. Provisional Patent Application No. 63/313,358, filed Feb. 24, 2022, the entirety of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] Embodiments of the disclosure relate generally to memory sub-systems, and more specifically, relate to selective increase and decrease to pass voltages for programing a memory device.

BACKGROUND

[0003] A memory sub-system can include one or more memory devices that store data. The memory devices can be, for example, non-volatile memory devices and volatile memory devices. In general, a host system can utilize a memory sub-system to store data at the memory devices and to retrieve data from the memory devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various embodiments of the disclosure.

[0005] FIG. 1A illustrates an example computing system that includes a memory sub-system in accordance with some embodiments of the present disclosure.

[0006] FIG. 1B is a block diagram of a memory device in communication with a memory sub-system controller of a memory sub-system, in accordance with some embodiments of the present disclosure.

[0007] FIGS. 2A-2C are schematics of portions of an array of memory cells as could be used in a memory of the type described with reference to FIG. 1B according to an embodiment.

[0008] FIG. 3A is a schematic of a memory array and channel at the end of a program verify in accordance with some embodiments of the present disclosure.

[0009] FIG. 3B is a schematic of a memory array and channel during a pass voltage and a program voltage ramp up in accordance with some embodiments of the present disclosure.

[0010] FIG. 4 is a diagram of a simplified example of drain-to-source programming of a selected wordline of multiple wordlines of a memory array in accordance with some embodiments of the present disclosure.

[0011] FIG. 5 is a graph of an example voltage ramp comparison between a pass voltage and a program voltage during a program operation in accordance with some embodiments.

[0012] FIG. 6 is a flow diagram of an example method of selectively increasing a pass voltage during an initial phase of a program operation and decreasing the pass voltage later during the program operation in accordance with at least some embodiments.

[0013] FIG. 7 is a graph of different pass voltages being compared between a pass voltage level and a program voltage level during a program operation in accordance with at least one embodiment.

[0014] FIG. 8 is a block diagram of an example computer system in which embodiments of the present disclosure can operate.

DETAILED DESCRIPTION

[0015] Aspects of the present disclosure are directed to selective increase and decrease to pass voltages for programming a memory device of a memory sub-system. A memory sub-system can be a storage device, a memory module, or a hybrid of a storage device and memory module. Examples of storage devices and memory modules are described below in conjunction with FIG. 1A. A host system can utilize a memory sub-system that includes one or more components, such as memory devices that store data. The host system can provide data to be stored at the memory sub-system and can request data to be retrieved from the memory sub-system.

[0016] A memory sub-system can include high density non-volatile memory devices where retention of data is desired when no power is supplied to the memory device. For example, negative AND (NAND) memory, such as 3D flash NAND memory, offers storage in the form of compact, high density configurations. A non-volatile memory device is a package of one or more dice, each including one or more planes. For some types of non-volatile memory devices (e.g., NAND memory), each plane includes of a set of physical blocks. Each block includes of a set of pages. Each page includes of a set of memory cells (“cells”). A cell is an electronic circuit that stores information. A data block hereinafter refers to a unit of the memory device used to store data and can include a group of memory cells, a word line group, a word line, or individual memory cells. Each data block can include a number of sub-blocks, where each sub-block is defined by an associated pillar (e.g., a vertical conductive trace) extending from a shared bitline. Memory pages (also referred to herein as “pages”) store one or more bits of binary data corresponding to data received from the host system. To achieve high density, a string of memory cells in a non-volatile memory device can be constructed to include a number of memory cells at least partially surrounding a pillar of channel material. The memory cells can be coupled to access lines, which are commonly referred to as “word lines,” often fabricated in common with the memory cells, so as to form an array of strings in a block of memory (e.g., “memory array”). The compact nature of certain non-volatile memory devices, such as 3D flash NAND memory, means word lines are common to many memory cells within a block of memory.

[0017] During a programming operation, selected memory cell(s) can be programmed with the application of a programming voltage to a selected word line. Due to the word line being common to multiple memory cells, unselected memory cells can be subject to the same programming voltage as the selected memory cell(s). If not otherwise preconditioned, the unselected memory cells can experience effects from the programming voltage on the common word line. These programming voltage effects can include the condition of charge being stored in the unselected memory cells which are expected to maintain stored data. This programming voltage effect is termed a “programming disturbance” or “program disturb” effect. The program disturb effect can render the charge stored in the unselected memory cells unreadable altogether or, although still apparently readable, the contents of the memory cell can be read as a data value different than the intended data value stored before application of the programming voltage.

[0018] The presence of residue electrons, such as electrons trapped or otherwise remaining on a charge storage structure after an earlier operation (e.g., a program verify operation), can contribute to the program disturb effect. At the end of a program verify operation, for example, a pass voltage (V_{pass}) applied to the word lines that are not being programmed ramps down and word lines on the source side having a high threshold voltage will cut off prior to word lines with a lower threshold voltage. Therefore, electrons will be trapped in the channel region (e.g., the pillar) of array transistors connected to the word lines with the lower threshold voltage (e.g., between the cut off word lines) and become residue electrons. Since the channel region (e.g., the pillar) in some non-volatile memory devices is a floating channel that may not be connected to a bulk grounded body, there is generally no path for residual electrons in the channel region to discharge other than

towards the source of the memory string.

[0019] These residue electrons can contribute to program disturb in a number of ways. Regular data word lines (e.g., word lines lower down the string) can suffer from hot-electron (“hot-e”) disturb where a large voltage differential between the gate and source causes the residue electrons to be injected from a drain depletion region into the floating gate. This hot-e disturb can permanently change switching characteristics of the transistor associated with the selected memory cell, which is an undesired outcome. Although electrons are referred to herein as causing hot-e disturb, holes can also cause hot-e disturb in other cases, and thus, this disclosure is equally applicable to preventing hot-e disturb from injecting holes as well. In addition, the top few word lines in the string might suffer from insufficient boosting when the channel material of unselected memory cells is at a voltage sufficiently different than the programming voltage. This difference in voltage can initiate an electrostatic field of sufficient magnitude to change the charge on a word line and cause the contents of the memory cell to be programmed inadvertently or read incorrectly.

[0020] Aspects of the present disclosure address the above and other deficiencies by increasing the pass voltage on multiple wordlines of the memory array (other than the selected wordline) during the initial phase of an inhibit program operation. An “inhibit” program operation makes reference to applying voltages to wordlines that prevent (or “inhibit”) connected memory cells from being programmed. Thus, the initial phase of an inhibit program is intended to move electrons out of the source side of the channel region of the memory array in a source-to-drain program operation and generally follows directly after a program verify operation having been performed on the memory array. By applying a higher pass voltage to the wordlines at the beginning of such a program operation (e.g., during the initial phase), any residue electrons (or holes) can be forced out of the source side of the channel region before the program voltage applied to the selected wordline is sufficiently increased to a level that would cause hot-e injection. This solution may be applicable to both source-to-drain and drain-to-source program operations of the memory array.

[0021] More specifically, control logic located on a memory device that is coupled with the memory array can perform some additional operations to mitigate the hot-e disturb. These operations can include, for example, detecting a program operation directed at a selected wordline of multiple wordlines of the memory array and determining, during an initial phase of the program operation, whether a program voltage being applied to the selected wordline satisfies a threshold program voltage. In at least some embodiments, this threshold voltage is between 20-50% of a maximum program voltage to eventually be applied to the selected wordline, as may be pre-designed or configured after manufacturing, for example. The operations can further include, in response to the program voltage not satisfying the threshold program voltage, adding a base offset voltage to an initial pass voltage to generate a higher pass voltage. In at least some embodiments, the initial pass voltage is a percentage of an initial program voltage, and thus, the higher pass voltage is greater than this percentage of the initial program voltage. The operation can further include causing the higher pass voltage to be applied to a remainder of the multiple wordlines other than the selected wordline.

[0022] In these embodiments, as programming continues, the programming voltage increases until surpassing the threshold program voltage. Further, the higher pass voltage grows at a first increase rate during this initial phase. In at least some embodiments, the operations performed by the control logic further include detecting, during programming associated with the selected wordline, that the program voltage satisfies the threshold program voltage. The operations can further include causing, in response to the detecting, the higher pass voltage to be reduced to a lower pass voltage, e.g., which is the initial pass voltage growing at a second increase rate. In some embodiments, the first increase rate is different than the second increase rate. For example the first increase rate can be zero while the second increase rate is positive, or the first increase rate can be lower than the second increase rate.

[0023] Advantages of this approach include, but are not limited to, mitigation or prevention of hot-

e injection and disturb associated with an inhibit program operation performed on a memory array after that memory array has undergone a previous program verify operation. Further, by reducing the pass voltages on the unselected wordlines after the initial increase to the higher pass voltage, the pass voltages do not reach a maximum pass voltage too early, thus preventing degradation (e.g., narrowing) of a read margin (E0) between a first voltage distribution and a first read level as a result of programming the selected memory cell. Additional advantages would be apparent to one skilled in the art in possession of the following disclosure.

[0024] FIG. 1A illustrates an example computing system **100** that includes a memory sub-system **110** in accordance with some embodiments of the present disclosure. The memory sub-system **110** can include media, such as one or more volatile memory devices (e.g., memory device **140**), one or more non-volatile memory devices (e.g., memory device **130**), or a combination of such media or memory devices.

[0025] The memory device **130** can be made up of bits arranged in a two-dimensional or three-dimensional grid, also referred to as a memory array. Memory cells are formed onto a silicon wafer in an array of columns (also hereinafter referred to as bitlines) and rows (also hereinafter referred to as wordlines). A wordline can refer to one or more rows of memory cells of a memory device that are used with one or more bitlines to generate the address of each of the memory cells. The intersection of a bitline and wordline constitutes the address of the memory cell.

[0026] A memory sub-system **110** can be a storage device, a memory module, or a combination of a storage device and memory module. Examples of a storage device include a solid-state drive (SSD), a flash drive, a universal serial bus (USB) flash drive, an embedded Multi-Media Controller (eMMC) drive, a Universal Flash Storage (UFS) drive, a secure digital (SD) card, and a hard disk drive (HDD). Examples of memory modules include a dual in-line memory module (DIMM), a small outline DIMM (SO-DIMM), and various types of non-volatile dual in-line memory modules (NVDIMMs).

[0027] The computing system **100** can be a computing device such as a desktop computer, laptop computer, network server, mobile device, a vehicle (e.g., airplane, drone, train, automobile, or other conveyance), Internet of Things (IoT) enabled device, embedded computer (e.g., one included in a vehicle, industrial equipment, or a networked commercial device), or such computing device that includes memory and a processing device.

[0028] The computing system **100** can include a host system **120** that is coupled to one or more memory sub-systems **110**. In some embodiments, the host system **120** is coupled to multiple memory sub-systems **110** of different types. FIG. 1A illustrates one example of a host system **120** coupled to one memory sub-system **110**. The host system **120** can provide data to be stored at the memory sub-system **110** and can request data to be retrieved from the memory sub-system **110**. As used herein, “coupled to” or “coupled with” generally refers to a connection between components, which can be an indirect communicative connection or direct communicative connection (e.g., without intervening components), whether wired or wireless, including connections such as electrical, optical, magnetic, etc.

[0029] The host system **120** can include a processor chipset and a software stack executed by the processor chipset. The processor chipset can include one or more cores, one or more caches, a memory controller (e.g., NVDIMM controller), and a storage protocol controller (e.g., PCIe controller, SATA controller). The host system **120** uses the memory sub-system **110**, for example, to write data to the memory sub-system **110** and read data from the memory sub-system **110**.

[0030] The host system **120** can be coupled to the memory sub-system **110** via a physical host interface. Examples of a physical host interface include, but are not limited to, a serial advanced technology attachment (SATA) interface, a peripheral component interconnect express (PCIe) interface, universal serial bus (USB) interface, Fibre Channel, Serial Attached SCSI (SAS), a double data rate (DDR) memory bus, Small Computer System Interface (SCSI), a dual in-line memory module (DIMM) interface (e.g., DIMM socket interface that supports Double Data Rate

(DDR)), etc. The physical host interface can be used to transmit data between the host system **120** and the memory sub-system **110**. The host system **120** can further utilize an NVM Express (NVMe) interface to access components (e.g., memory devices **130**) when the memory sub-system **110** is coupled with the host system **120** by the physical host interface (e.g., PCIe bus). The physical host interface can provide an interface for passing control, address, data, and other signals between the memory sub-system **110** and the host system **120**. FIG. 1A illustrates a memory sub-system **110** as an example. In general, the host system **120** can access multiple memory sub-systems via a same communication connection, multiple separate communication connections, and/or a combination of communication connections.

[0031] The memory devices **130**, **140** can include any combination of the different types of non-volatile memory devices and/or volatile memory devices. The volatile memory devices (e.g., memory device **140**) can be, but are not limited to, random access memory (RAM), such as dynamic random access memory (DRAM) and synchronous dynamic random access memory (SDRAM).

[0032] Some examples of non-volatile memory devices (e.g., memory device **130**) include a negative-and (NAND) type flash memory and write-in-place memory, such as a three-dimensional cross-point (“3D cross-point”) memory device, which is a cross-point array of non-volatile memory cells. A cross-point array of non-volatile memory cells can perform bit storage based on a change of bulk resistance, in conjunction with a stackable cross-gridded data access array. Additionally, in contrast to many flash-based memories, cross-point non-volatile memory can perform a write in-place operation, where a non-volatile memory cell can be programmed without the non-volatile memory cell being previously erased. NAND type flash memory includes, for example, two-dimensional NAND (2D NAND) and three-dimensional NAND (3D NAND).

[0033] Each of the memory devices **130** can include one or more arrays of memory cells. One type of memory cell, for example, single-level cells (SLC) can store one bit per cell. Other types of memory cells, such as multi-level cells (MLCs), triple-level cells (TLCs), quad-level cells (QLCs), and penta-level cells (PLCs) can store multiple bits per cell. In some embodiments, each of the memory devices **130** can include one or more arrays of memory cells such as SLCs, MLCs, TLCs, QLCs, PLCs or any combination of such. In some embodiments, a particular memory device can include an SLC portion, and an MLC portion, a TLC portion, a QLC portion, or a PLC portion of memory cells. The memory cells of the memory devices **130** can be grouped as pages that can refer to a logical unit of the memory device used to store data. With some types of memory (e.g., NAND), pages can be grouped to form blocks.

[0034] Although non-volatile memory components such as a 3D cross-point array of non-volatile memory cells and NAND type flash memory (e.g., 2D NAND, 3D NAND) are described, the memory device **130** can be based on any other type of non-volatile memory, such as read-only memory (ROM), phase change memory (PCM), self-selecting memory, other chalcogenide based memories, ferroelectric transistor random-access memory (FeTRAM), ferroelectric random access memory (FeRAM), magneto random access memory (MRAM), Spin Transfer Torque (STT)-MRAM, conductive bridging RAM (CBRAM), resistive random access memory (RRAM), oxide based RRAM (OxRAM), negative-or (NOR) flash memory, or electrically erasable programmable read-only memory (EEPROM).

[0035] A memory sub-system controller **115** (or controller **115** for simplicity) can communicate with the memory devices **130** to perform operations such as reading data, writing data, or erasing data at the memory devices **130** and other such operations. The memory sub-system controller **115** can include hardware such as one or more integrated circuits and/or discrete components, a buffer memory, or a combination thereof. The hardware can include a digital circuitry with dedicated (i.e., hard-coded) logic to perform the operations described herein. The memory sub-system controller **115** can be a microcontroller, special purpose logic circuitry (e.g., a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), etc.), or other suitable processor.

[0036] The memory sub-system controller **115** can include a processing device, which includes one or more processors (e.g., processor **117**), configured to execute instructions stored in a local memory **119**. In the illustrated example, the local memory **119** of the memory sub-system controller **115** includes an embedded memory configured to store instructions for performing various processes, operations, logic flows, and routines that control operation of the memory sub-system **110**, including handling communications between the memory sub-system **110** and the host system **120**.

[0037] In some embodiments, the local memory **119** can include memory registers storing memory pointers, fetched data, etc. The local memory **119** can also include read-only memory (ROM) for storing micro-code. While the example memory sub-system **110** in FIG. **1A** has been illustrated as including the memory sub-system controller **115**, in another embodiment of the present disclosure, a memory sub-system **110** does not include a memory sub-system controller **115**, and can instead rely upon external control (e.g., provided by an external host, or by a processor or controller separate from the memory sub-system).

[0038] In general, the memory sub-system controller **115** can receive commands or operations from the host system **120** and can convert the commands or operations into instructions or appropriate commands to achieve the desired access to the memory devices **130**. The memory sub-system controller **115** can be responsible for other operations such as wear leveling operations, garbage collection operations, error detection and error-correcting code (ECC) operations, encryption operations, caching operations, and address translations between a logical address (e.g., a logical block address (LBA), namespace) and a physical address (e.g., physical block address) that are associated with the memory devices **130**. The memory sub-system controller **115** can further include host interface circuitry to communicate with the host system **120** via the physical host interface. The host interface circuitry can convert the commands received from the host system into command instructions to access the memory devices **130** as well as convert responses associated with the memory devices **130** into information for the host system **120**.

[0039] The memory sub-system **110** can also include additional circuitry or components that are not illustrated. In some embodiments, the memory sub-system **110** can include a cache or buffer (e.g., DRAM) and address circuitry (e.g., a row decoder and a column decoder) that can receive an address from the memory sub-system controller **115** and decode the address to access the memory devices **130**.

[0040] In some embodiments, the memory devices **130** include local media controllers **135** that operate in conjunction with memory sub-system controller **115** to execute operations on one or more memory cells of the memory devices **130**. An external controller (e.g., memory sub-system controller **115**) can externally manage a memory device **130** (e.g., perform media management operations on the memory device **130**). In some embodiments, memory sub-system **110** is a managed memory device, which is a raw memory device **130** having control logic (e.g., local media controller **135**) on the die and a controller (e.g., memory sub-system controller **115**) for media management within the same memory device package. An example of a managed memory device is a managed NAND (MNAND) device. Memory device **130**, for example, can represent a single die or multiple dice having some control logic (e.g., local media controller **135**) embodied thereon. In some embodiments, one or more components of the memory sub-system **110** are omitted.

[0041] In one embodiment, memory sub-system **110** includes a memory interface component **113**. Memory interface component **113** is responsible for handling interactions of memory sub-system controller **115** with the memory devices of memory sub-system **110**, such as memory device **130**. For example, memory interface component **113** can send memory access commands corresponding to requests received from host system **120** to memory device **130**, such as program commands, read commands, or other commands. In addition, memory interface component **113** can receive data from memory device **130**, such as data retrieved in response to a read command or a confirmation

that a program command was successfully performed. In some embodiments, the memory sub-system controller **115** includes at least a portion of the memory interface **113**. For example, the memory sub-system controller **115** can include a processor **117** (e.g., a processing device) configured to execute instructions stored in local memory **119** for performing the operations described herein. In some embodiments, the memory interface component **113** is part of the host system **110**, an application, or an operating system.

[0042] In some embodiments, memory device **130** includes the local media controller **135**, a memory array **104**, and a page buffer **152** coupled with the memory array **104**. The local media controller **135**, also referred to as control logic herein, can direct the amount and timing of the increases and decreases to the pass voltages during various programming operations as described in detail herein, e.g., to mitigate hot-electron disturb. In some embodiments, the page buffer **152** provides the circuitry used to program data to the memory cells of the memory device **130** and to read the data out of the memory cells. This circuitry includes, for example, a number of latches for buffering data. In some embodiments, this control logic is integrated in whole or in part within the memory sub-system controller **115** and/or the host system **120**. Further details with regards to the operations of local media controller **135** are described below.

[0043] FIG. **1B** is a simplified block diagram of a first apparatus, in the form of a memory device **130**, in communication with a second apparatus, in the form of a memory sub-system controller **115** of a memory sub-system (e.g., memory sub-system **110** of FIG. **1A**), according to an embodiment. Some examples of electronic systems include personal computers, personal digital assistants (PDAs), digital cameras, digital media players, digital recorders, games, appliances, vehicles, wireless devices, mobile telephones and the like. The memory sub-system controller **115** (e.g., a controller external to the memory device **130**), may be a memory controller or other external host device.

[0044] Memory device **130** includes an array of memory cells **104** logically arranged in rows and columns. Memory cells of a logical row are typically connected to the same access line (e.g., a wordline) while memory cells of a logical column are typically selectively connected to the same data line (e.g., a bitline). A single access line may be associated with more than one logical row of memory cells and a single data line may be associated with more than one logical column. Memory cells (not shown in FIG. **1B**) of at least a portion of array of memory cells **104** are capable of being programmed to one of at least two target data states.

[0045] Row decode circuitry **108** and column decode circuitry **109** are provided to decode address signals. Address signals are received and decoded to access the array of memory cells **104**. Memory device **130** also includes input/output (I/O) control circuitry **160** to manage input of commands, addresses and data to the memory device **130** as well as output of data and status information from the memory device **130**. An address register **114** is in communication with I/O control circuitry **160** and row decode circuitry **108** and column decode circuitry **109** to latch the address signals prior to decoding. A command register **124** is in communication with I/O control circuitry **160** and local media controller **135** to latch incoming commands.

[0046] A controller (e.g., the local media controller **135** internal to the memory device **130**) controls access to the array of memory cells **104** in response to the commands and generates status information for the external memory sub-system controller **115**, i.e., the local media controller **135** is configured to perform access operations (e.g., read operations, programming operations and/or erase operations) on the array of memory cells **104**. The local media controller **135** is in communication with row decode circuitry **108** and column decode circuitry **109** to control the row decode circuitry **108** and column decode circuitry **109** in response to the addresses. In one embodiment, local media controller **135** can perform a ganged SLC verify operation to concurrently verify (i.e., at least partially overlapping in time) that memory cells in two separate sub-blocks of a block of memory array **104** were successfully programmed with data from two separate pages using a single sensing operation.

[0047] The local media controller **135** is also in communication with a cache register **172**. Cache register **172** latches data, either incoming or outgoing, as directed by the local media controller **135** to temporarily store data while the array of memory cells **104** is busy writing or reading, respectively, other data. During a program operation (e.g., write operation), data may be passed from the cache register **172** to the data register **170** for transfer to the array of memory cells **104**; then new data may be latched in the cache register **172** from the I/O control circuitry **160**. During a read operation, data may be passed from the cache register **172** to the I/O control circuitry **160** for output to the memory sub-system controller **115**; then new data may be passed from the data register **170** to the cache register **172**. The cache register **172** and/or the data register **170** may form (e.g., may form a portion of) the page buffer **152** of the memory device **130**. The page buffer **152** may further include sensing devices (not shown in FIG. **1B**) to sense a data state of a memory cell of the array of memory cells **104**, e.g., by sensing a state of a data line connected to that memory cell. A status register **122** may be in communication with I/O control circuitry **160** and the local memory controller **135** to latch the status information for output to the memory sub-system controller **115**.

[0048] Memory device **130** receives control signals at the memory sub-system controller **115** from the local media controller **135** over a control link **132**. For example, the control signals can include a chip enable signal CE #, a command latch enable signal CLE, an address latch enable signal ALE, a write enable signal WE #, a read enable signal RE #, and a write protect signal WP #. Additional or alternative control signals (not shown) may be further received over control link **132** depending upon the nature of the memory device **130**. In one embodiment, memory device **130** receives command signals (which represent commands), address signals (which represent addresses), and data signals (which represent data) from the memory sub-system controller **115** over a multiplexed input/output (I/O) bus **134** and outputs data to the memory sub-system controller **115** over I/O bus **134**.

[0049] For example, the commands may be received over input/output (I/O) pins [7:0] of I/O bus **134** at I/O control circuitry **160** and may then be written into command register **124**. The addresses may be received over input/output (I/O) pins [7:0] of I/O bus **134** at I/O control circuitry **160** and may then be written into address register **114**. The data may be received over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device at I/O control circuitry **160** and then may be written into cache register **172**. The data may be subsequently written into data register **170** for programming the array of memory cells **104**.

[0050] In an embodiment, cache register **172** may be omitted, and the data may be written directly into data register **170**. Data may also be output over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device. Although reference may be made to I/O pins, they may include any conductive node providing for electrical connection to the memory device **130** by an external device (e.g., the memory sub-system controller **115**), such as conductive pads or conductive bumps as are commonly used.

[0051] It will be appreciated by those skilled in the art that additional circuitry and signals can be provided, and that the memory device **130** of FIG. **1B** has been simplified. It should be recognized that the functionality of the various block components described with reference to FIG. **1B** may not necessarily be segregated to distinct components or component portions of an integrated circuit device. For example, a single component or component portion of an integrated circuit device could be adapted to perform the functionality of more than one block component of FIG. **1B**. Alternatively, one or more components or component portions of an integrated circuit device could be combined to perform the functionality of a single block component of FIG. **1B**. Additionally, while specific I/O pins are described in accordance with popular conventions for receipt and output of the various signals, it is noted that other combinations or numbers of I/O pins (or other I/O node structures) may be used in the various embodiments.

[0052] FIG. **2A-2C** are schematics of portions of an array of memory cells **200A**, such as a NAND

memory array, as could be used in a memory of the type described with reference to FIG. 1B according to an embodiment, e.g., as a portion of the array of memory cells **104**. Memory array **200A** includes access lines, such as wordlines **202.sub.0** to **202.sub.N**, and data lines, such as bitlines **204.sub.0** to **204.sub.M**. The wordlines **202** can be connected to global access lines (e.g., global wordlines), not shown in FIG. 2A, in a many-to-one relationship. For some embodiments, memory array **200A** can be formed over a semiconductor that, for example, can be conductively doped to have a conductivity type, such as a p-type conductivity, e.g., to form a p-well, or an n-type conductivity, e.g., to form an n-well.

[0053] Memory array **200A** can be arranged in rows (each corresponding to a wordline **202**) and columns (each corresponding to a bitline **204**). Each column can include a string of series-connected memory cells (e.g., non-volatile memory cells), such as one of NAND strings **206.sub.0** to **206.sub.M**. Each NAND string **206** can be connected (e.g., selectively connected) to a common source (SRC) **216** and can include memory cells **208.sub.0** to **208.sub.N**. The memory cells **208** can represent non-volatile memory cells for storage of data. The memory cells **208** of each NAND string **206** can be connected in series between a select gate **210** (e.g., a field-effect transistor), such as one of the select gates **210.sub.0** to **210.sub.M** (e.g., that can be source select transistors, commonly referred to as select gate source), and a select gate **212** (e.g., a field-effect transistor), such as one of the select gates **212.sub.0** to **212.sub.M** (e.g., that can be drain select transistors, commonly referred to as select gate drain). Select gates **210.sub.0** to **210.sub.M** can be commonly connected to a select line **214**, such as a source select line (SGS), and select gates **212.sub.0** to **212.sub.M** can be commonly connected to a select line **215**, such as a drain select line (SGD). Although depicted as traditional field-effect transistors, the select gates **210** and **212** can utilize a structure similar to (e.g., the same as) the memory cells **208**. The select gates **210** and **212** can represent a number of select gates connected in series, with each select gate in series configured to receive a same or independent control signal.

[0054] A source of each select gate **210** can be connected to common source **216**. The drain of each select gate **210** can be connected to a memory cell **208.sub.0** of the corresponding NAND string **206**. For example, the drain of select gate **210.sub.0** can be connected to memory cell **208.sub.0** of the corresponding NAND string **206.sub.0**. Therefore, each select gate **210** can be configured to selectively connect a corresponding NAND string **206** to the common source **216**. A control gate of each select gate **210** can be connected to the select line **214**.

[0055] In some embodiments, the drain select line (SGD) and source select line (SGS) are separated and biased to different sources or to a commonly shared source. Further, the SGD and SGS lines can be ones of several layers of memory sub-blocks, each layer that can include various connections as explained above. Further, the common source **216** (SRC) can be a plate that can be fully or partially segmented or may be full connected.

[0056] The drain of each select gate **212** can be connected to the bitline **204** for the corresponding NAND string **206**. For example, the drain of select gate **212.sub.0** can be connected to the bitline **204.sub.0** for the corresponding NAND string **206.sub.0**. The source of each select gate **212** can be connected to a memory cell **208.sub.N** of the corresponding NAND string **206**. For example, the source of select gate **212.sub.0** can be connected to memory cell **208.sub.N** of the corresponding NAND string **206.sub.0**. Therefore, each select gate **212** can be configured to selectively connect a corresponding NAND string **206** to the corresponding bitline **204**. A control gate of each select gate **212** can be connected to select line **215**.

[0057] The memory array **200A** in FIG. 2A can be a quasi-two-dimensional memory array and can have a generally planar structure, e.g., where the common source **216**, NAND strings **206** and bitlines **204** extend in substantially parallel planes. Alternatively, the memory array **200A** in FIG. 2A can be a three-dimensional memory array, e.g., where NAND strings **206** can extend substantially perpendicular to a plane containing the common source **216** and to a plane containing the bitlines **204** that can be substantially parallel to the plane containing the common source **216**.

[0058] Typical construction of memory cells **208** includes a data-storage structure **234** (e.g., a floating gate, charge trap, and the like) that can determine a data state of the memory cell (e.g., through changes in threshold voltage), and a control gate **236**, as shown in FIG. **2A**. The data-storage structure **234** can include both conductive and dielectric structures while the control gate **236** is generally formed of one or more conductive materials. In some cases, memory cells **208** can further have a defined source/drain (e.g., source) **230** and a defined source/drain (e.g., drain) **232**. The memory cells **208** have their control gates **236** connected to (and in some cases form) a wordline **202**.

[0059] A column of the memory cells **208** can be a NAND string **206** or a number of NAND strings **206** selectively connected to a given bitline **204**. A row of the memory cells **208** can be memory cells **208** commonly connected to a given wordline **202**. A row of memory cells **208** can, but need not, include all the memory cells **208** commonly connected to a given wordline **202**. Rows of the memory cells **208** can often be divided into one or more groups of physical pages of memory cells **208**, and physical pages of the memory cells **208** often include every other memory cell **208** commonly connected to a given wordline **202**. For example, the memory cells **208** commonly connected to wordline **202.sub.N** and selectively connected to even bitlines **204** (e.g., bitlines **204.sub.0**, **204.sub.2**, **204.sub.4**, etc.) can be one physical page of the memory cells **208** (e.g., even memory cells) while memory cells **208** commonly connected to wordline **202.sub.N** and selectively connected to odd bitlines **204** (e.g., bitlines **204.sub.1**, **204.sub.3**, **204.sub.5**, etc.) can be another physical page of the memory cells **208** (e.g., odd memory cells).

[0060] Although bitlines **204.sub.3-204.sub.5** are not explicitly depicted in FIG. **2A**, it is apparent from the figure that the bitlines **204** of the array of memory cells **200A** can be numbered consecutively from bitline **204.sub.0** to bitline **204.sub.M**. Other groupings of the memory cells **208** commonly connected to a given wordline **202** can also define a physical page of memory cells **208**. For certain memory devices, all memory cells commonly connected to a given wordline can be deemed a physical page of memory cells. The portion of a physical page of memory cells (which, in some embodiments, could still be the entire row) that is read during a single read operation or programmed during a single programming operation (e.g., an upper or lower page of memory cells) can be deemed a logical page of memory cells. A block of memory cells can include those memory cells that are configured to be erased together, such as all memory cells connected to wordlines **202.sub.0-202.sub.N** (e.g., all NAND strings **206** sharing common wordlines **202**).

Unless expressly distinguished, a reference to a page of memory cells herein refers to the memory cells of a logical page of memory cells. Although the example of FIG. **2A** is discussed in conjunction with NAND flash, the embodiments and concepts described herein are not limited to a particular array architecture or structure, and can include other structures (e.g., SONOS, phase change, ferroelectric, etc.) and other architectures (e.g., AND arrays, NOR arrays, etc.).

[0061] FIG. **2B** is another schematic of a portion of an array of memory cells **200B** as could be used in a memory of the type described with reference to FIG. **1B**, e.g., as a portion of the array of memory cells **104**. Like numbered elements in FIG. **2B** correspond to the description as provided with respect to FIG. **2A**. FIG. **2B** provides additional detail of one example of a three-dimensional NAND memory array structure. The three-dimensional NAND memory array **200B** can incorporate vertical structures which can include semiconductor pillars where a portion of a pillar can act as a channel region of the memory cells of NAND strings **206**. The NAND strings **206** can be each selectively connected to a bitline **204.sub.0-204.sub.M** by a select transistor **212** (e.g., that can be drain select transistors, commonly referred to as select gate drain) and to a common source **216** by a select transistor **210** (e.g., that can be source select transistors, commonly referred to as select gate source). Multiple NAND strings **206** can be selectively connected to the same bitline **204**. Subsets of NAND strings **206** can be connected to their respective bitlines **204** by biasing the select lines **215.sub.0-215.sub.K** to selectively activate particular select transistors **212** each between a NAND string **206** and a bitline **204**. The select transistors **210** can be activated by biasing the select

line **214**. Each wordline **202** can be connected to multiple rows of memory cells of the memory array **200B**. Rows of memory cells that are commonly connected to each other by a particular wordline **202** can collectively be referred to as tiers.

[0062] FIG. **2C** is a further schematic of a portion of an array of memory cells **200C** as could be used in a memory of the type described with reference to FIG. **1B**, e.g., as a portion of the array of memory cells **104**. Like numbered elements in FIG. **2C** correspond to the description as provided with respect to FIG. **2A**. The array of memory cells **200C** can include strings of series-connected memory cells (e.g., NAND strings) **206**, access (e.g., word) lines **202**, data (e.g., bit) lines **204**, select lines **214** (e.g., source select lines), select lines **215** (e.g., drain select lines) and a source **216** as depicted in FIG. **2A**. A portion of the array of memory cells **200A** can be a portion of the array of memory cells **200C**, for example.

[0063] FIG. **2C** depicts groupings of NAND strings **206** into blocks of memory cells **250**, e.g., blocks of memory cells **250.sub.0-250.sub.L**. Blocks of memory cells **250** can be groupings of memory cells **208** that can be erased together in a single erase operation, sometimes referred to as erase blocks. Each block of memory cells **250** can represent those NAND strings **206** commonly associated with a single select line **215**, e.g., select line **215.sub.0**. The source **216** for the block of memory cells **250.sub.0** can be a same source as the source **216** for the block of memory cells **250.sub.L**. For example, each block of memory cells **250.sub.0-250.sub.L** can be commonly selectively connected to the source **216**. Access lines **202** and select lines **214** and **215** of one block of memory cells **250** can have no direct connection to access lines **202** and select lines **214** and **215**, respectively, of any other block of memory cells of the blocks of memory cells **250.sub.0-250.sub.L**.

[0064] The bitlines **204.sub.0-204.sub.M** can be connected (e.g., selectively connected) to a buffer portion **240**, which can be a portion of the page buffer of the memory device **130**. The buffer portion **240** can correspond to a memory plane (e.g., the set of blocks of memory cells **250.sub.0-250.sub.L**). The buffer portion **240** can include sense circuits (which can include sense amplifiers) for sensing data values indicated on respective bitlines **204**.

[0065] FIG. **3A** is a schematic of a memory array **300** and a channel **305** at the end of a program verify in accordance with some embodiments of the present disclosure. To turn off the wordlines (WL) at the end of the program verify, the wordline voltages are taken from an asserted voltage (and example of which is illustrated as 8V) to an approximately zero voltage ("0V"). Because the threshold voltages (V_t) of the memory cell transistors below the selected memory cell can be high while concurrently applying a zero bias to the wordlines, a variation in voltage across the channel **305** can often cause residual electrons **310** to be trapped inside the channel **305** at the end of the program verify operation.

[0066] FIG. **3B** is a schematic of the memory array **300** and the channel **305** during a pass voltage and a program voltage ramp up in accordance with some embodiments of the present disclosure. These trapped residual electrons **310** can cause hot-e injection/disturb at a gate of the selected wordline (WL_n) during a program inhibit that is performed during an initial phase of a new program operation to the memory array. For example, the electrons **310** in the channel can be accelerated into the gate of a selected memory cell that is being programmed, causing hot carrier injection, also known as hot-e disturb. This acceleration is generally caused by a relatively strong vertical electric field generated by the increase in wordline voltage to the program voltage (V_{pgm}), which redirects the residual electrons from escaping to being injected at the gate of the selected memory cell. This hot-e disturb can permanently change switching characteristics of the transistor associated with the selected memory cell, an undesired outcome. The hot-e disturb issue can also be exacerbated at higher device temperatures. Although electrons are referred to herein as causing hot-e disturb, holes can also cause hot-e disturb in other cases, and thus, this disclosure is equally applicable to preventing hot-e disturb from injected holes as well.

[0067] FIG. **4** is a diagram of a simplified example of drain-to-source programming of a selected

wordline of multiple wordlines of a memory array **400** in accordance with some embodiments of the present disclosure. As illustrated, the pass voltage (V_{pass}) is applied to unselected wordlines to the drain side of the selected memory cell. As the V_{pass} voltage is applied to these wordlines, the channel is gradually turned on. A program voltage (V_{pgm}) is applied to the selected wordline (WL_n) of the selected memory cell. The bitline voltage (V_{bl}) is illustrated as being applied to this memory array **400**, e.g., which can be a string of memory cells as discussed with reference to FIGS. 2A-2C.

[0068] The X below the L15 memory cell illustrates how a lower V_{pass} voltage might cause the bitline voltage to not fully conduct down the channel, risking not being able to program the selected wordline (WL_n). In such drain-to-source programming, a higher pass voltage at the beginning (or initial phase) of programming enables the source side to be fully conducted, thus facilitating the program of the selected wordline (WL_n).

[0069] FIG. 5 is a graph of an example voltage ramp comparison between a pass voltage and a program voltage during a program operation in accordance with some embodiments. Along the x-axis of the graph is illustrated a program voltage (V_{pgm}) applied to a selected wordline during an inhibit program operation. Along the y-axis is illustrated the pass voltage (V_{pass}) to be applied to the remainder of the multiple wordlines other than selected wordline. An initial pass voltage **502** is generally to be higher than the program voltage by a certain percentage, e.g., referred to as a pass voltage ratio (V_{pass.sub.ratio}). Thus, the initial pass voltage **502** can be set as the start program voltage multiplied by the pass voltage ratio, or V_{pgm.sub.start}*V_{pass.sub.ratio}. A pass voltage **514** is illustrated that begins at this initial pass voltage level and grows (e.g., ramps) at an initial increase rate governed by a step ratio (Step.sub.ratio).

[0070] According to at least some embodiments, the control logic, during an initial phase of the program operation, determines whether a program voltage being applied to the selected wordline satisfies a threshold program voltage **520**. The difference in threshold program voltage and an initial program voltage (e.g., V_{pgm} threshold **520**–V_{pgm_start}) can be set to between 20-50% of a difference between a maximum program voltage and the initial program voltage (e.g., V_{pgm_max}–V_{pgm_start}) to be applied to the selected wordline, which is also sometimes referred to as a target program voltage. The control logic can then add, in response to the program voltage not satisfying the threshold program voltage, a base offset voltage **530** to the initial pass voltage **502** to generate a higher pass voltage **504**, the initial pass voltage being a percentage of an initial program voltage. Thus, as the program operation starts, the higher pass voltage **504**, which is growing (or ramping) at a subsequent increase rate, is applied to a remainder of the multiple wordlines other than the selected wordline.

[0071] In at least some embodiments, as programming progresses, the control logic detects that the program voltage satisfies the threshold program voltage **520**, e.g., has reached the threshold program voltage **520** at the selected wordline. In response, the control logic causes, in response to the detecting, the higher pass voltage to be reduced to a lower pass voltage, e.g., incurring a voltage drop **510** back down to the initial pass voltage **514**, which is growing at the initial increase rate. In some embodiments, the subsequent increase rate is different than the initial increase rate. In at least one embodiment, the subsequent increase rate is zero while the initial increase rate is positive.

[0072] FIG. 6 is a flow diagram of an example method **600** of selectively increasing a pass voltage during an initial phase of a program operation and decreasing the pass voltage later during the program operation in accordance with at least some embodiments. The method **600** can be performed by processing logic that can include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method **600** is performed by local media controller **135** of FIG. 1A and FIG. 1B. Although shown in a particular sequence or order, unless otherwise specified, the order of the processes can be modified. Thus, the illustrated embodiments should be understood only as

examples, and the illustrated processes can be performed in a different order, and some processes can be performed in parallel. Additionally, one or more processes can be omitted in various embodiments. Thus, not all processes are required in every embodiment. Other process flows are possible.

[0073] At operation **610**, a program operation is detected. More specifically, the processing logic detects a program operation directed at a selected wordline of multiple wordlines of a memory array. For example, the selected wordline can be one of the wordlines **202.sub.0** to **202.sub.N** in the array of memory cells **200A** (FIG. 2A) or the array of memory cells **200B** (FIG. 2B).

[0074] At operation **620**, a program voltage is checked. More specifically, the processing logic determines, during an initial phase of the program operation, whether the program voltage being applied to the selected wordline satisfies a threshold program voltage. In at least some embodiments, satisfying this threshold voltage includes a difference between the threshold program voltage and the initial program voltage reaching between 20-50% of a difference between a maximum program voltage and the initial program voltage to be applied to the selected wordline. If yes, the program voltage is satisfied, then at operation **630**, the processing logic employs an initial pass voltage growing at an initial increase rate, and no offset voltage may be required.

[0075] If, however, the program voltage does not satisfy the threshold program voltage, at operation **640**, a higher pass voltage is employed. More specifically, in response to the program voltage not satisfying the threshold program voltage at operation **620**, the processing logic can add a base offset voltage to an initial pass voltage to generate a higher pass voltage, the initial pass voltage being a percentage of an initial program voltage applied to the selected wordline.

[0076] At operation **650**, the higher pass voltage is applied. More specifically, the processing logic causes the higher pass voltage to be applied to a remainder of the multiple wordlines other than the selected wordline. This higher pass voltage can grow at a first increase rate. As described herein, the higher pass voltage encourages the exit of residual electrons in the channel of the memory array, thus mitigating against experiencing hot-electron injection or disturb at the selected memory cell.

[0077] At operation **660**, the program voltage is checked again. More specifically, the processing logic determines, during programming associated with the selected wordline, whether the program voltage satisfies the threshold program voltage. If no, then the processing logic loops back to operation **650** and continues to apply the higher pass voltage.

[0078] If, however, the program voltage does not satisfy the threshold program voltage, at operation **670**, the higher pass voltage is reduced. More specifically, the processing logic causes, in response to detecting the program voltage satisfy the threshold program voltage, the higher pass voltage to be reduced to a lower pass voltage that is the initial pass voltage growing at a second increase rate. In at least some embodiment, the first increase rate is different than the second increase rate. In at least one embodiment, the first increase rate is zero and the second increase rate is positive.

[0079] FIG. 7 is a graph of different pass voltages being compared between a pass voltage level and a program voltage level during a program operation in accordance with at least one embodiment. This graph illustrates how multiple different higher pass voltages can be generated, including that are associated with V_{pass} , V_{pass1} that is lower than V_{pass} , and V_{pass2} that is higher than V_{pass} . In these embodiments, the V_{pass1} pass voltage can be applied to one or more wordlines on the erase side of a selected wordline (WLn) and the V_{pass2} pass voltage can be applied to one or more wordlines on the programmed side of the selected wordline (WLn). Further, the V_{pass} pass voltage is applied to a remainder of the wordlines other than the selected wordline and these one or more wordlines on the erase side and on the programmed side of the selected wordline (WLn).

[0080] More specifically, with additional reference to FIG. 6, at operation **640**, the processing logic would instead add a base offset voltage to an initial pass voltage, to an initial first pass voltage, and to an initial second pass voltage, to respectively generate a higher pass voltage ($V_{pass'}$), a higher

first pass voltage (V_{pass1}), and a higher second pass voltage (V_{pass2}). Then, at operation **650**, the processing logic would cause the higher pass voltage to be applied to a subset of the plurality of wordlines other than the selected wordline; the higher first pass voltage to be applied to one or more wordlines that are adjacent to an erase side of the selected wordline, the higher first pass voltage being less than the higher pass voltage; and the higher second pass voltage to be applied to one or more wordlines that is adjacent to a programmed side of the selected wordline, the higher second pass voltage being greater than the higher pass voltage.

[0081] In these embodiments, the higher pass voltage, the higher first pass voltage, and the higher second pass voltage can each have a first increase rate. Then, at operation **660**, the processing logic would detect, during programming associated with the selected wordline, that the program voltage satisfies the threshold program voltage. And, at operation **670**, the processing logic would cause, in response to the detecting, the higher pass voltage, the higher first pass voltage, and the higher second pass voltage to each be reduced to a respective lower pass voltage that increases at a second increase rate. In some embodiments, the first increase rate is different than the second increase rate. In at least one embodiment, the first increase rate is zero and the second increase rate is positive. Further, in at least one embodiment, the first increase rate for each respective pass voltage level can vary to some degree and/or the second increase rate for each respective pass voltage level can vary to some degree.

[0082] FIG. **8** illustrates an example machine of a computer system **800** within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, can be executed. In some embodiments, the computer system **800** can correspond to a host system (e.g., the host system **120** of FIG. **1A**) that includes, is coupled to, or utilizes a memory sub-system (e.g., the memory sub-system **110** of FIG. **1A**) or can be used to perform the operations of a controller (e.g., to execute an operating system to perform operations corresponding to the local media controller **135** of FIGS. **1A-1B**). In alternative embodiments, the machine can be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, and/or the Internet. The machine can operate in the capacity of a server or a client machine in client-server network environment, as a peer machine in a peer-to-peer (or distributed) network environment, or as a server or a client machine in a cloud computing infrastructure or environment.

[0083] The machine can be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, a switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

[0084] The example computer system **800** includes a processing device **802**, a main memory **804** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory **806** (e.g., flash memory, static random access memory (SRAM), etc.), and a data storage system **818**, which communicate with each other via a bus **830**.

[0085] Processing device **802** represents one or more general-purpose processing devices such as a microprocessor, a central processing unit, or the like. More particularly, the processing device can be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device **802** can also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device **802** is configured to execute instructions **826** for performing the operations and steps discussed herein. The computer

system **800** can further include a network interface device **808** to communicate over the network **820**.

[0086] The data storage system **818** can include a machine-readable storage medium **824** (also known as a computer-readable medium) on which is stored one or more sets of instructions **826** or software embodying any one or more of the methodologies or functions described herein. The instructions **826** can also reside, completely or at least partially, within the main memory **804** and/or within the processing device **802** during execution thereof by the computer system **800**, the main memory **804** and the processing device **802** also constituting machine-readable storage media. The machine-readable storage medium **824**, data storage system **818**, and/or main memory **804** can correspond to the memory sub-system **110** of FIG. **1A**.

[0087] In one embodiment, the instructions **826** include instructions to implement functionality corresponding to the local media controller **135** of FIGS. **1A-1B**). While the machine-readable storage medium **824** is shown in an example embodiment to be a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media that store the one or more sets of instructions. The term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present disclosure. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, and magnetic media.

[0088] Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

[0089] It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. The present disclosure can refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage systems.

[0090] The present disclosure also relates to an apparatus for performing the operations herein. This apparatus can be specially constructed for the intended purposes, or it can include a general purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program can be stored in a computer readable storage medium, such as, but not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

[0091] The algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various general purpose systems can be used with programs in accordance with the teachings herein, or it can prove convenient to construct a more specialized apparatus to perform the method. The structure for a variety of these systems will appear as set forth in the description below. In addition, the present disclosure is not described with reference to

any particular programming language. It will be appreciated that a variety of programming languages can be used to implement the teachings of the disclosure as described herein.

[0092] The present disclosure can be provided as a computer program product, or software, that can include a machine-readable medium having stored thereon instructions, which can be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). In some embodiments, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as a read only memory (“ROM”), random access memory (“RAM”), magnetic disk storage media, optical storage media, flash memory components, etc.

[0093] In the foregoing specification, embodiments of the disclosure have been described with reference to specific example embodiments thereof. It will be evident that various modifications can be made thereto without departing from the broader spirit and scope of embodiments of the disclosure as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A memory device comprising: a memory array comprising a plurality of wordlines; and control logic operatively coupled with the memory array, the control logic to perform operations comprising: determining, during an initial phase of a program operation, whether a program voltage being applied to a selected wordline of the memory array satisfies a threshold program voltage; increasing, in response to the program voltage not satisfying the threshold program voltage, an initial pass voltage to generate a higher pass voltage; and causing the higher pass voltage to be applied to unselected wordlines of the plurality of wordlines before programming one or more memory cells associated with the selected wordline.
2. The memory device of claim 1, wherein the initial phase is of an inhibit program operation, which is intended to move electrons out of a source side of a channel of the memory array, and wherein increasing the initial pass voltage comprises adding a base offset voltage to the initial pass voltage.
3. The memory device of claim 1, wherein the higher pass voltage grows at a first increase rate, and wherein the operations further comprise: detecting, during programming associated with the selected wordline, that the program voltage satisfies the threshold program voltage; and causing, in response to the detecting, the higher pass voltage to be reduced to a lower pass voltage comprising the initial pass voltage growing at a second increase rate.
4. The memory device of claim 3, wherein the first increase rate is different than the second increase rate.
5. The memory device of claim 3, wherein the first increase rate is zero and the second increase rate is positive.
6. The memory device of claim 1, wherein the program operation follows directly after a program verify operation having been performed on the memory array.
7. The memory device of claim 1, wherein the program operation is a drain-to-source program operation of the memory array.
8. The memory device of claim 1, wherein a difference between the threshold program voltage and the initial program voltage is between 20-50% of a difference between a maximum program voltage and the initial program voltage to be applied to the selected wordline.
9. A memory device comprising: a memory array comprising a plurality of wordlines; and control logic operatively coupled with the memory array, the control logic to perform operations comprising: determining, during an initial phase of a program operation, that a program voltage being applied to a selected wordline of the memory array does not satisfy a threshold program

voltage; increasing an initial first pass voltage and an initial second pass voltage to respectively generate a higher first pass voltage and a higher second pass voltage, wherein the higher second pass voltage is higher than the higher first pass voltage; causing the higher first pass voltage to be applied to one or more wordlines that are adjacent to an erase side of the selected wordline; and causing the higher second pass voltage to be applied to one or more wordlines that are adjacent to a programmed side of the selected wordline.

10. The memory device of claim 9, wherein increasing the initial first pass voltage and the initial second pass voltage comprises adding a base offset voltage to the initial first pass voltage and to the initial second pass voltage.

11. The memory device of claim 9, wherein the operations further comprise: increasing an initial pass voltage to a higher pass voltage; and causing the higher pass voltage to be applied to a subset of the plurality of wordlines other than the selected wordline, wherein the higher first pass voltage is less than the higher pass voltage and the higher second pass voltage is greater than the higher pass voltage.

12. The memory device of claim 11, wherein the higher pass voltage, the higher first pass voltage, and the higher second pass voltage each have a first increase rate, and wherein the operations further comprise: detecting, during programming associated with the selected wordline, that the program voltage satisfies the threshold program voltage; and causing, in response to the detecting, the higher pass voltage, the higher first pass voltage, and the higher second pass voltage to each be reduced to a respective lower pass voltage that increases at a second increase rate.

13. The memory device of claim 12, wherein the first increase rate is different than the second increase rate.

14. The memory device of claim 12, wherein the first increase rate is zero and the second increase rate is positive.

15. A memory device comprising: a memory array comprising a plurality of wordlines; and control logic operatively coupled with the memory array, the control logic to perform operations comprising: determining, during an initial phase of a program operation, that a program voltage being applied to a selected wordline of the memory array does not satisfy a threshold program voltage, wherein the initial phase is of an inhibit program operation, which is intended to move electrons out of a source side of a channel of the memory array; and in response to the determining: increasing an initial first pass voltage and an initial second pass voltage to respectively generate a higher first pass voltage and a higher second pass voltage, wherein the higher second pass voltage is higher than the higher first pass voltage; causing the higher first pass voltage to be applied to one or more wordlines that are adjacent to an erase side of the selected wordline; and causing the higher second pass voltage to be applied to one or more wordlines that are adjacent to a programmed side of the selected wordline.

16. The memory device of claim 15, wherein increasing the initial first pass voltage and the initial second pass voltage comprises adding a base offset voltage to the initial first pass voltage and to the initial second pass voltage.

17. The memory device of claim 15, wherein the operations further comprise: increasing an initial pass voltage to a higher pass voltage; and causing the higher pass voltage to be applied to a subset of the plurality of wordlines other than the selected wordline, wherein the higher first pass voltage is less than the higher pass voltage and the higher second pass voltage is greater than the higher pass voltage.

18. The memory device of claim 17, wherein the higher pass voltage, the higher first pass voltage, and the higher second pass voltage each have a first increase rate, and wherein the operations further comprise: detecting, during programming associated with the selected wordline, that the program voltage satisfies the threshold program voltage; and causing, in response to the detecting, the higher pass voltage, the higher first pass voltage, and the higher second pass voltage to each be reduced to a respective lower pass voltage that increases at a second increase rate.

19. The memory device of claim 18, wherein the first increase rate is different than the second increase rate.

20. The memory device of claim 18, wherein the first increase rate is zero and the second increase rate is positive.
